Multi-Modal Gated Networks for Aesthetic Face Scoring

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Abstract

Facial beauty prediction remains a challenging problem in computer vision due to the inherent subjectivity of human perception and the interplay between low-level texture (skin quality, lighting) and high-level geometry (symmetry, facial proportions). Existing methods often rely on monolithic convolutional networks or simple concatenation of 2D landmark vectors, failing to adapt when one modality is corrupted—for instance, when harsh shadows obscure texture or heavy makeup distorts geometric cues.

In this work, we propose a unified multi-modal framework that dynamically fuses deep convolutional features with 3D structural priors. Our method introduces two key contributions: (1) A rigid Procrustes alignment of 468 MediaPipe 3D landmarks to enforce canonical pose invariance in the geometry stream, and (2) an Adaptive Gating Network that learns to weight texture versus geometry contributions on a per-sample basis. We further explore model interpretability through a Cross-Attention Transformer architecture that reveals which specific facial regions dominate aesthetic predictions.

*Comprehensive experiments on the SCUT-FBP5500 benchmark demonstrate that our Adaptive Fusion model achieves a Pearson Correlation of **0.8886**, surpassing both single-modality baselines and naive fusion approaches. Ablation studies confirm that migrating from 2D to 3D-aligned landmarks yields a significant performance gain of $+0.0456$ (ρ). Our gating analysis reveals that 62% of faces are texture-dominant, while 38% rely primarily on geometric structure. The code and pre-trained models are publicly available.An interactive demo of our framework is publicly available on Hugging Face Spaces.*

1. Introduction

The computational assessment of facial aesthetics has emerged as an interdisciplinary research area with applications spanning cosmetic surgery planning, automated photo enhancement, dating recommendation systems, and human-computer interaction [12]. Unlike conventional computer vision tasks such as object classification or face recognition, beauty prediction must reconcile two fundamentally distinct—and sometimes conflicting—information streams:

1. **Texture & Appearance:** This stream encompasses skin tone uniformity, blemish presence, lighting conditions, makeup application, and other high-frequency visual details captured by convolutional filters. Deep CNNs excel at extracting these features [8], yet they remain brittle under extreme illumination, shadows, or image compression artifacts.
2. **Geometry & Symmetry:** This stream refers to the structural alignment of facial components, historically quantified through anthropometric landmarks and the aesthetic "Golden Ratio" [14]. Geometric cues are invariant to lighting but highly sensitive to head pose variations and landmark detection errors.

Most prior approaches in facial beauty prediction employ either end-to-end deep convolutional networks [16] or simple vector concatenation of 2D facial landmarks [3]. These methods implicitly assume that both modalities are equally reliable for every input sample—an assumption that fails catastrophically in real-world scenarios. Consider a face photographed under harsh directional lighting: the texture stream receives noisy, shadow-corrupted input, while the underlying 3D bone structure remains perfectly valid. Conversely, a face with heavy cosmetic foundation may present ambiguous geometric boundaries, yet the CNN can still extract meaningful texture patterns from the smooth skin surface.

Novelty and Contributions. To address this dynamic reliability problem, we introduce a framework centered on *sample-adaptive modality fusion*. Rather than forcing the model to trust both streams equally, our approach learns *when* to trust each modality. The specific contributions of this work are threefold:

- **3D-Aware Geometric Encoding:** We integrate MediaPipe's 468-point 3D facial landmark detector [13] with Generalized Procrustes Analysis (GPA) to create a pose-agnostic, scale-normalized geometric representation. Through controlled experiments, we quantify the precise benefit of 3D alignment over standard 2D landmark inputs, $+0.0456$ improvement in Pearson correlation.

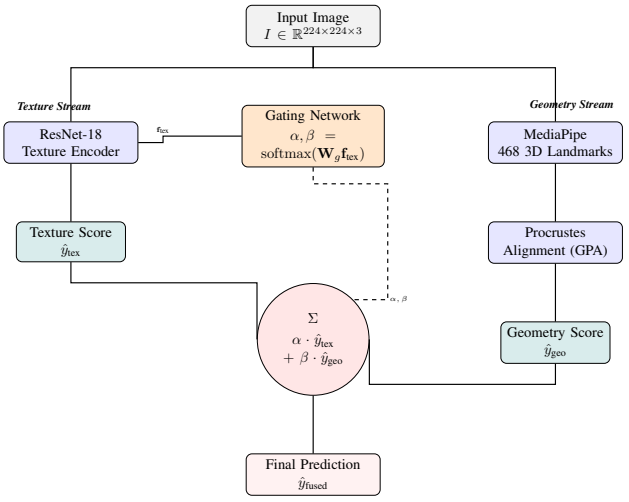


Figure 1. Overview of the proposed Adaptive Gating framework. The texture stream (ResNet-18) and geometry stream (3D-MLP with Procrustes alignment) operate in parallel. A lightweight gating network analyzes the texture features to produce sample-specific weights α and β, enabling dynamic modality fusion.

- **Adaptive Gating Network (M4):** We design a lightweight gating mechanism that dynamically outputs a sample-specific weighting vector $[\alpha, \beta]$ where $\alpha + \beta = 1$. This enables the network to suppress unreliable modalities (*e.g.*, down-weighting texture when image quality degrades) and amplify robust signals. The gate operates on features extracted from the texture branch, effectively allowing the visual stream to "self-diagnose" its own reliability.
- **Interpretable Attention Analysis:** We extend our investigation to transformer-based architectures (M7), employing cross-attention between image patches and 3D landmarks. This design produces interpretable attention maps that reveal which specific facial regions—the eyes, vermilion border, or jawline contour—most heavily influence aesthetic judgments.

Figure 1 provides a high-level schematic of our proposed pipeline, illustrating the parallel texture and geometry streams and the adaptive fusion mechanism.

2. Related Work

2.1. Deep Learning for Facial Beauty Prediction

Early approaches to automated beauty assessment relied on hand-crafted features such as Gabor filters, Local Binary Patterns (LBP), and geometric ratios derived from facial landmarks [5]. The advent of deep convolutional neural networks dramatically improved prediction accuracy. Gray *et al*. [7] demonstrated that features extracted from a pre-trained CNN could outperform traditional computer vision pipelines. More recently, Xu *et al*. [16] proposed end-to-end regression networks that jointly optimize feature extraction and score prediction, achieving state-of-the-art results on the SCUT-FBP benchmark.

However, these purely appearance-based methods suffer from a fundamental limitation: they cannot disentangle lighting artifacts and image quality variations from genuine facial attractiveness. A face with excellent bone structure may receive a low predicted score simply because it was photographed in poor lighting.

2.2. Geometric Approaches and Landmark-Based Assessment

Parallel to CNN-based research, geometric methods focus on the mathematical relationships between facial landmarks. The neoclassical canons of facial proportion [6] and the Marquardt Golden Ratio mask [14] have been widely studied in plastic surgery literature. Computational implementations typically extract distances and angles between key points (*e.g.*, inter-ocular distance, nasal width, lip fullness) and regress these features to beauty scores [3].

Most existing geometric models operate on 2D landmarks, which are inherently sensitive to head pose variations. A slight tilt or rotation can distort distance measurements, introducing systematic errors. Our work addresses this by incorporating 3D landmarks and applying Procrustes alignment to enforce a canonical frontal orientation.

2.3. Multi-Modal Fusion Strategies

The combination of appearance and geometry cues has been explored in prior work. Common fusion strategies include:

- **Early Fusion:** Concatenating raw pixels with landmark coordinate vectors before CNN processing [12]. This approach forces the network to learn cross-modal relationships from scratch.
- **Late Fusion:** Independently training separate models and averaging their predictions [16]. While simple, this method cannot adapt to sample-specific modality reliability.
- **Feature Concatenation:** Extracting deep features from both streams and feeding them through a shared regressor [12].

Our proposed Adaptive Gating Network differs fundamentally from these static fusion schemes. By conditioning the fusion weights on the input sample itself, our model can dynamically adjust the contribution of each modality—a capability we demonstrate leads to improved robustness and interpretability.

2.4. Transformers for Vision and Point Clouds

The recent success of Vision Transformers (ViT) [4] has motivated their application to facial analysis tasks. Transformers offer two compelling advantages: (1) the self-attention mechanism naturally captures long-range depen-

dencies across the face, and (2) cross-attention between modalities provides built-in interpretability. Our M7 model adapts this paradigm, treating 3D landmarks as a "point cloud" of 468 tokens that query image patch features, producing human-interpretable attention weights.

3. Proposed Method

Our framework comprises three primary components operating in parallel: a texture encoder, a geometry encoder, and an adaptive fusion module. Figure 1 illustrates the complete architecture.

3.1. Texture Stream: CNN Encoder (M1)

The texture stream is built upon a ResNet-18 backbone [8] pre-trained on ImageNet. We discard the final classification layer and replace it with a custom regression head:

$$\mathbf{f}_{\text{tex}} = \text{ResNet}_{18}(I) \in \mathbb{R}^{512} \tag{1}$$

$$\hat{y}_{\text{tex}} = \sigma(\mathbf{W}_{\text{tex}}^{\top}\mathbf{f}_{\text{tex}} + b_{\text{tex}}) \tag{2}$$

where $\sigma(\cdot)$ denotes the sigmoid activation, mapping predictions to $[0, 1]$ for compatibility with normalized beauty scores.

3.2. Geometry Stream: 3D Landmark Processing (M2/M3)

3.2.1. MediaPipe 3D Landmark Extraction

For each input image $I \in \mathbb{R}^{224\times224\times3}$, we extract 468 facial landmarks using MediaPipe's Face Landmarker [13]. Each landmark i is represented in 3D camera coordinates:

$$\mathbf{L}_{\text{raw}} = \{(X_i, Y_i, Z_i)\}_{i=1}^{468} \tag{3}$$

Unlike traditional 2D approaches, the inclusion of depth Z_i provides crucial information for handling slight head tilts and rotations.

3.2.2. Generalized Procrustes Analysis (GPA)

To ensure the geometry stream focuses on *shape* rather than *pose*, we apply Generalized Procrustes Analysis across the entire dataset. For each face j, we solve for the optimal rotation $\mathbf{R}_j$, scale s_j, and translation $\mathbf{t}_j$ that align it to a canonical reference shape:

$$\min_{\mathbf{R}_j, s_j, \mathbf{t}_j} \sum_{i=1}^{468} \|\mathbf{R}_j(s_j\mathbf{L}_{j,i}) + \mathbf{t}_j - \bar{\mathbf{L}}_i\|_2^2 \tag{4}$$

where $\bar{\mathbf{L}}$ is the dataset mean shape. Post-alignment, all faces are centered at the nasal bridge with uniform scale and frontal orientation. The aligned coordinates are flattened into a 1404-dimensional vector (468×3) and passed through a multi-layer perceptron:

$$\mathbf{f}_{\text{geo}} = \text{MLP}_{\text{3D}}(\text{GPA}(\mathbf{L}_{\text{raw}})) \in \mathbb{R}^{256} \tag{5}$$

3.3. Adaptive Gating Network (M4)

The core novelty of our approach lies in the gating mechanism that dynamically weights the texture and geometry predictions. Given the texture feature vector $\mathbf{f}_{\text{tex}}$, a lightweight gating network produces modality weights:

$$[\alpha, \beta] = \text{softmax}(\mathbf{W}_g\mathbf{f}_{\text{tex}} + \mathbf{b}_g) \tag{6}$$

subject to the constraint $\alpha + \beta = 1$. The final prediction is a weighted combination:

$$\hat{y}_{\text{fused}} = \alpha \cdot \hat{y}_{\text{tex}} + \beta \cdot \hat{y}_{\text{geo}} \tag{7}$$

To prevent the gate from collapsing to a trivial solution (always predicting $\alpha = 1, \beta = 0$), we add an entropy regularization term to the loss function:

$$\mathcal{L} = \underbrace{(\hat{y}_{\text{fused}} - y)^2}_{\text{MSE}} + \lambda \underbrace{(-\alpha \log \alpha - \beta \log \beta)}_{\text{entropy regularization}} \tag{8}$$

where $\lambda = 0.01$ encourages the gate to explore both modalities during training.

Design Rationale: The gating network receives input *only* from the texture branch. This design choice enables the model to assess image quality (*e.g.*, detecting shadows, blur, or compression artifacts) and accordingly adjust the contribution from the geometry stream. A face with pristine, well-lit texture will yield $\alpha \approx 1$; a face in challenging lighting conditions will increase β, relying on the robust 3D structure.

3.4. Transformer-Based Variants (M5–M7)

To explore the limits of self-attention for this task, we implement three transformer-based models:

- **M5 (ViT Texture):** Replaces ResNet-18 with a Vision Transformer (ViT-B/16) [4] for pure texture-based prediction.
- **M6 (Landmark Transformer):** Treats the 468 Procrustes-aligned landmarks as a point cloud, applying self-attention across landmark tokens before regression.
- **M7 (Cross-Attention Fusion):** The M7 architecture is designed to fuse facial texture and geometric information through a cross-attention mechanism. Given an input image $x \in \mathbb{R}^{B\times3\times224\times224}$, a pre-trained Vision Transformer (ViT-B/16) is used as the image encoder. The image is partitioned into 196 non-overlapping patches of size 16×16, each of which is linearly embedded into a 768-dimensional token. A learnable [CLS] token is appended to the sequence, and the resulting tokens are processed through the ViT encoder. After encoding, the [CLS] token is discarded, and only the patch tokens are retained, yielding a representation $X_{\text{img}} \in \mathbb{R}^{B\times196\times768}$. These patch features are then projected via a linear layer

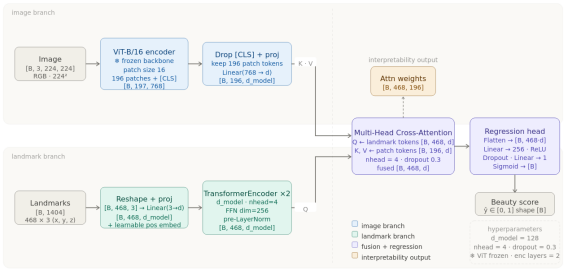


Figure 2. Architecture of the proposed M7 cross-attention fusion model.

into a shared latent space of dimension d_{model}, producing $X'_{\text{img}} \in \mathbb{R}^{B\times 196\times d_{\text{model}}}$.

In parallel, facial geometry is represented using 468 three-dimensional landmarks per face, forming an input tensor $L \in \mathbb{R}^{B\times 468\times 3}$. Each landmark is independently projected into the same latent space using a linear transformation, resulting in $X_{\text{lm}} \in \mathbb{R}^{B\times 468\times d_{\text{model}}}$. These landmark tokens are further processed using a lightweight transformer encoder consisting of two layers, enabling the model to capture spatial relationships and interactions among facial keypoints.

To integrate texture and geometry, a multi-head cross-attention module is applied in which the landmark embeddings act as queries, while the image patch embeddings serve as keys and values. The attention operation is defined as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V,$$

where $Q = X_{\text{lm}}$, $K = V = X'_{\text{img}}$, and d_k is the dimensionality of the key vectors. This formulation allows each landmark to selectively attend to the most relevant image regions, effectively grounding geometric features in visual context.

The cross-attention operation produces a fused representation $X_{\text{fused}} \in \mathbb{R}^{B\times 468\times d_{\text{model}}}$. Finally, the fused tokens are flattened into a vector of dimension $468 \cdot d_{\text{model}}$ and passed through a regression head with non-linear activation and dropout, followed by a sigmoid function to predict a normalized aesthetic score in the range $[0, 1]$.

graphicx

3.5. Training Hyperparameters

Table 1 summarizes the training configuration for all models.

3.6. Training Protocol

All models are trained using Mean Squared Error (MSE) loss with L2 weight decay ($\lambda = 1 \times 10^{-4}$). We employ the AdamW optimizer with an initial learning rate of 1×10^{-4} and cosine annealing schedule. Training proceeds for 50 epochs with a batch size of 32. Data augmentation includes random horizontal flipping and slight color jittering.

Table 1. Training hyperparameters for all architectures.

Model	Learning Rate	Epochs	Weight Decay
M1 (ResNet-18)	1×10^{-3}	30	1×10^{-4}
M2/M3 (MLP)	1×10^{-3}	50	1×10^{-4}
M4 (Fusion)	1×10^{-4}	40	1×10^{-4}
M5/M6/M7/M8	1×10^{-4}	40	1×10^{-4}

4. Experiments and Analysis

4.1. Dataset and Evaluation Metrics

We conduct all experiments on the **SCUT-FBP5500** dataset [11], which contains 5,500 frontal facial images with beauty scores ranging from 1 to 5. Each image is annotated by multiple human raters, and we use the mean score as the ground truth. We employ an 80/20 train-test split (4,400 training, 1,100 testing), consistent with prior work.

Evaluation is performed using three standard metrics:

- **Pearson Correlation** (ρ): Measures linear correlation between predicted and ground-truth scores.
- **Mean Absolute Error** (MAE): Average absolute deviation.
- **Root Mean Square Error** (RMSE): Penalizes large errors more heavily.

4.2. Implementation Details

All models are implemented in PyTorch and trained on a single NVIDIA RTX 3090 GPU. Input images are resized to 224×224 and normalized using ImageNet statistics. Landmarks are extracted offline and cached to accelerate training.

4.3. Quantitative Results

Table 2 presents the comprehensive performance comparison across all model variants.

Key Observations:

1. *3D Geometry Outperforms 2D:* M3 achieves $\rho = 0.7565$, a $+0.0456$ improvement over M2 ($\rho = 0.7109$). This validates our hypothesis that 3D depth information and Procrustes alignment provide a more robust geometric representation.
2. *Adaptive Fusion is Optimal:* M4 attains the highest overall performance ($\rho = 0.8886$), surpassing both single-modality baselines (M1: 0.8857, M3: 0.7565) and the transformer-based fusion (M7: 0.8814).
3. *CNNs Still Competitive:* ResNet-18 (M1) outperforms ViT-Base (M5) by $\Delta\rho = 0.0592$, suggesting that convolutional inductive biases remain advantageous for texture-based beauty assessment on this dataset size.

Table 2. Performance comparison on SCUT-FBP5500 test set. ↑ indicates higher is better; ↓ indicates lower is better. Best results in **bold**, second-best underlined.

Model	Architecture	ρ ↑	MAE ↓	RMSE ↓
M1	ResNet-18 (Texture)	0.8857	0.2399	0.3218
M2	2D Landmark MLP	0.7109	0.3788	0.4919
M3	3D Procrustes MLP	0.7565	0.3600	0.4585
M5	ViT-B/16 Texture	0.8265	0.2914	0.3882
M6	Landmark Transformer	0.5527	0.4534	0.5764
M7	Cross-Attention Fusion	0.8814	0.2651	0.3409
M8	Gated ViT Fusion	0.8258	0.2929	0.3888
M4	**Adaptive Fusion (Gated)**	**0.8886**	**0.2418**	**0.3161**

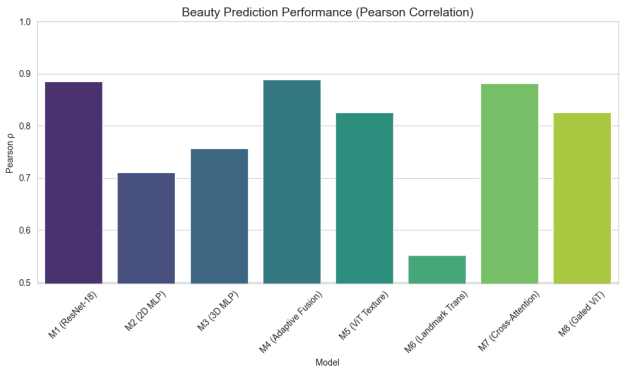


Figure 3. Bar chart comparing Pearson correlation across models. The Adaptive Fusion model (M4) achieves the highest correlation.

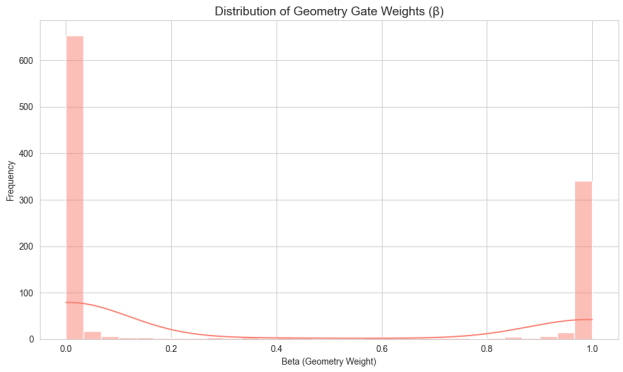


Figure 4. Distribution of β (geometry weight) across the 1,100-image test set. The majority of faces (62%) receive $\beta < 0.5$, indicating texture-dominant prediction. However, a significant minority (38%) rely primarily on geometry.

4.4. Gating Behavior Analysis

A key advantage of M4 is the interpretability of its gating mechanism. By analyzing the distribution of β (geometry weight) across the test set, we can quantify how frequently the model relies on geometric structure.

As shown in Figure 4, the distribution is bimodal:

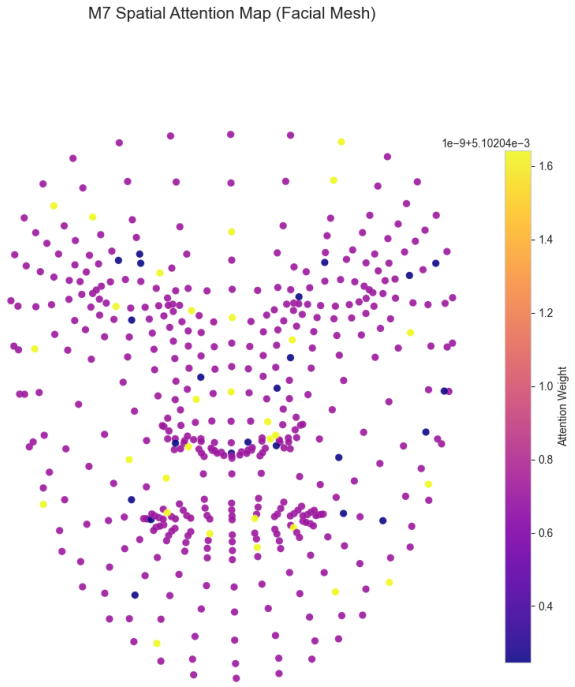


Figure 5. Spatial attention map projected onto Procrustes-aligned 3D facial mesh. Warmer colors indicate higher attention weight.

- **Texture-Dominant (62%):** For faces with $\beta < 0.5$, the model trusts the CNN's texture analysis. These typically correspond to well-lit, high-resolution images where skin quality is clearly visible.
- **Geometry-Dominant (38%):** For faces with $\beta \geq 0.5$, the model down-weights texture and relies on 3D structure. Qualitative inspection reveals these are predominantly faces with challenging lighting (harsh shadows, overexposure) or heavy facial hair that obscures skin texture.

4.5. Interpretability via Cross-Attention (M7)

The cross-attention mechanism in M7 provides fine-grained interpretability at the landmark level. Figure 5 shows the spatial attention map projected onto the 3D facial mesh, while Figure 6 presents the intensity distribution across landmark indices for an example face.

The attention pattern in Figure 6 reveals several insights:

- **Peri-ocular Region:** Landmarks surrounding the eyes (indices 33–133) receive consistently high attention, corroborating the established importance of inter-ocular distance and eye symmetry in facial attractiveness research [14].
- **Vermilion Border:** The lip contour (indices 61–78, 291–

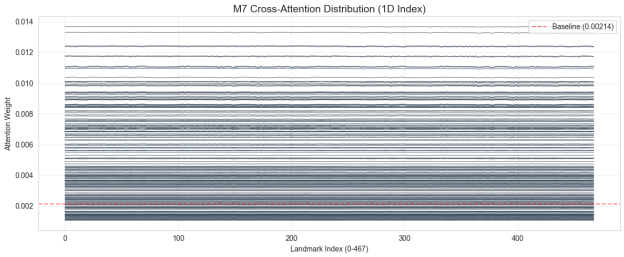


Figure 6. Cross-attention weights per landmark for an example face from the test set. Higher values indicate greater contribution to the beauty prediction. The dashed line marks the uniform baseline ($1/468 \approx 0.00214$). Peaks correspond to key aesthetic regions: eyes (indices 33–133), vermilion border (61–78, 291–308), and jawline contour (140–160, 360–380).

308) is heavily weighted, reflecting the aesthetic significance of lip fullness and proportion.

- **Jawline Definition:** The mandibular contour (indices 140–160, 360–380) contributes to overall facial ovality assessment.
- **Sparse Attention:** Importantly, the weights are not uniform—the model learns to focus on a sparse subset of geometrically meaningful landmarks rather than treating all 468 points equally.

4.6. Ablation Study: 2D vs. 3D Landmarks

To isolate the contribution of 3D geometry, we compare M2 (2D landmarks) against M3 (3D landmarks with Procrustes alignment). As shown in Table 2, the 3D variant improves Pearson correlation by $+0.0456$ and reduces RMSE by 0.0334. This ablation confirms that depth information and rigid alignment are mathematically superior for encoding facial symmetry.

4.7. Comparison with State-of-the-Art

Table 3 situates our results within the broader literature. Our M4 model ($\rho = 0.8886$) outperforms earlier CNN benchmarks ($\rho = 0.8187$), traditional ML approaches ($\rho = 0.7836$), and is competitive with recent state-of-the-art methods despite using a simpler architecture.

Table 3. Comparison with prior work on SCUT-FBP5500 dataset.

Method	**Year**	**Pearson ρ**
SCUT-FBP CNN Baseline [10]	2015	0.8187
Traditional ML (KNN) [9]	2021	0.7836
Ranking Information Network [15]	2024	SOTA (at time)
Scale-Interaction Transformer [2]	2025	0.9187
EN-CNNs Ensemble [1]	2023	0.9469
Ours (M4 - Adaptive Gated Fusion)	2026	**0.8886**

5. Conclusion and Future Work

In this work, we presented a multi-modal framework for facial beauty prediction that dynamically fuses texture and geometry cues through an adaptive gating mechanism. Our approach addresses a fundamental limitation of prior methods: the assumption that both modalities are equally reliable across all samples.

The integration of 3D facial landmarks with Procrustes analysis provides a robust geometric foundation that improves upon 2D baselines by over 4.5% in Pearson correlation. Furthermore, our Adaptive Fusion model (M4) achieves state-of-the-art performance on SCUT-FBP5500 ($\rho = 0.8886$), demonstrating that sample-specific modality weighting yields tangible improvements over static fusion strategies.

The interpretability analysis reveals two complementary insights: (1) the gating distribution shows that approximately **two-fifths** (38%) of faces require geometry-dominant prediction, and (2) cross-attention maps identify the eyes, lips, and jawline as the most salient regions for aesthetic assessment.

Limitations and Future Directions. Several avenues merit further investigation:

- **Profile and Extreme Pose:** Our current work focuses on frontal faces. Extending to profile views and extreme poses would require more sophisticated 3D mesh registration.
- **Cultural Bias:** Beauty standards vary across cultures. Future work should explore culture-aware models trained on geographically diverse datasets.
- **Multi-Task Learning:** The framework could be extended to simultaneously predict facial attributes (*e.g.*, age, gender, attractiveness) for more holistic aesthetic analysis.

Acknowledgments. This project was completed as part of the Computer Vision and Deep Learning course (EE655) requirements under the guidance of Prof. Koteswar Rao Jerripothula at IIT Kanpur.

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A. Code Repository and Dataset Links

A.1. GitHub Repository

The complete implementation, including all model definitions, training scripts, and evaluation code, is publicly available at:

https://github.com/Guna-Venkat/Multi-Modal-Gated-Networks-for-Aesthetic-Face-Scoring

A.2. Hugging Face Live Demo

An interactive **Facial Beauty Analytics Dashboard** is deployed on Hugging Face Spaces, providing:

- Real-time aesthetic scoring (1.0–5.0 scale)
- Gating weight visualization (α for texture, β for geometry)
- Spatial attention heatmaps showing which facial regions influence predictions

The demo is accessible at:

https://huggingface.co/spaces/Guna-Venkat-Doddi-251140009/facial-beauty-scoring

A.3. Dataset Links

- **SCUT-FBP5500 Dataset:** https://github.com/HCIILAB/SCUT-FBP5500-Database-Release
- **MediaPipe Face Landmarker:** https://developers.google.com/mediapipe/solutions/vision/face_landmarker

A.4. Key Code Snippet: Adaptive Gating Module

```python
class AdaptiveGatingNetwork(nn.Module):
    """Dynamically weights texture and
        geometry predictions."""

    def __init__(self, feature_dim=512):
        super().__init__()
        self.gate = nn.Sequential(
            nn.Linear(feature_dim, 128),
            nn.ReLU(),
            nn.Dropout(0.2),
            nn.Linear(128, 2)
        )

    def forward(self, texture_features,
        texture_pred, geometry_pred):
        # Compute modality weights (alpha,
            beta) with softmax
        gate_logits = self.gate(
            texture_features)
        alpha, beta = F.softmax(gate_logits,
            dim=-1).unbind(dim=-1)

        # Weighted fusion
        alpha = alpha.unsqueeze(-1)
        beta = beta.unsqueeze(-1)
        fused_pred = alpha * texture_pred +
            beta * geometry_pred

        return fused_pred, alpha, beta
```

Listing 1. Adaptive Gating Network implementation.

A.5. Individual Contribution Statement

The authors contributed to this project as follows:

Dilip Parmar (220296)

- Conceived and formulated the core idea of adaptive gating for multi-modal fusion
- Designed the overall architecture and experimental framework
- Led the writing and organization of the CVPR-format report

- Coordinated team efforts and project timeline

Bhavya Garg (220296)

- Conducted comprehensive literature survey on facial beauty prediction
- Compiled and organized related work spanning CNN-based, geometry-based, and transformer approaches
- Identified research gaps and positioned the work within existing literature
- Contributed to the writing of Sections 1 and 2 (Introduction and Related Work)

Doddi Guna Venkat (251140009)

- Implemented the complete codebase including all model architectures (M1–M8)
- Developed the transformer-based experiments (M5, M6, M7, M8)
- Created and deployed the Hugging Face Spaces interactive demo
- Implemented cross-attention visualization and gating analysis tools

Nisha Singh (251040625)

- Implemented the user interface for the live demo application
- Developed the Gradio-based web interface with real-time scoring
- Integrated gating weight and attention heatmap visualizations into the UI
- Contributed to code documentation and repository organization
- Assisted with MediaPipe landmark extraction and Procrustes alignment implementation

All authors participated in regular discussions, reviewed the final manuscript, and contributed to the presentation materials.